



NETWORK NUTS

CONTAINERIZATION WITH DOCKER

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Containerization with Docker

OVERVIEW

Containers are a solution to the problem of how to get software to run reliably when moved from one computing environment to another.

Docker exploded onto the scene in 2013, and it's been causing excitement in IT circles ever since.

The application container technology provided by Docker promises to change the way that IT operations are carried out just as virtualization technology did a few years previously.

This course is best suited for system administrators, support engineers and anyone with Linux skills and now wanted to move into kubernetes or openshift cluster administration.

Where we are
TODAY

Course Outline

- Introduction to Microservices
 - Understanding different types of applications - monolithic / microservices
 - Comparison between virtual machines and containers
 - Introduction to container engines
 - Installing docker on Windows / Linux / AWS
 - Changing docker root dir
 - Relationship between container images & containers
 - Image registries
- Controlling Containers
 - Exploring container images
 - Best practices while running containers
 - Running containerized application
 - Accessing containerized application
 - Running container in detach mode
 - Accessing a running container
 - Relationship between container & process
 - Executing commands inside containers
 - Stopping & removing containers
 - Container restart policies
 - Understanding docker engine components



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Course Outline

- Configuring Persistent Storage
 - Understanding ephemeral & permanent storage
 - Exploring ephemeral storage
 - Creating docker volumes
 - Attaching containers to docker volumes
- Docker Networking
 - Understanding docker networking components
 - Understanding types of docker network - bridge, host, none, macvlan
 - Creating custom docker networks
 - Attaching container to custom networks
 - Connecting container to multiple networks
 - Exposing containers using port mapping
 - Removing networks
- Docker Compose
 - Understanding multi-container applications
 - Benefits & limitations of docker compose
 - Installing docker compose
 - Running multi-container applications using docker-compose.yaml
 - Managing multi-container applications with docker compose
- Creating custom images
 - Creating custom container images using docker commit
 - Understanding benefits of using Dockerfile
 - Creating custom images using Dockerfile method
 - Using custom images to run containers
 - Publishing custom images to image registries
 - Scanning images for vulnerabilities



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